

MPPT-CONTROLLED GRID-CONNECTED SOLAR PV SYSTEM

DISTRIBUTION AUTOMATION AND SMART GRID (EC9130)



Group	Group 11
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1. AIMS OF THE 14TH WEEK

This week focused on two core technical milestones to advance the MPPT-Controlled Grid-Connected Solar PV System towards full functionality:

1. Performance Evaluation – Validate overall system efficiency under varying irradiance and temperature conditions
2. Power Quality Check (THD) – Perform FFT-based THD analysis to verify grid harmonic distortion limits (THD < 5%)

2. SIMULINK MODEL OVERVIEW

The complete MATLAB/Simulink model implemented for this project is shown below. It integrates all subsystems: PV Array, Boost Converter with MPPT (P&O), DC-Link Voltage Control, 3-Phase Inverter, LCL Filter, PLL, abc-to-dq transformation, and Current Loop Control.

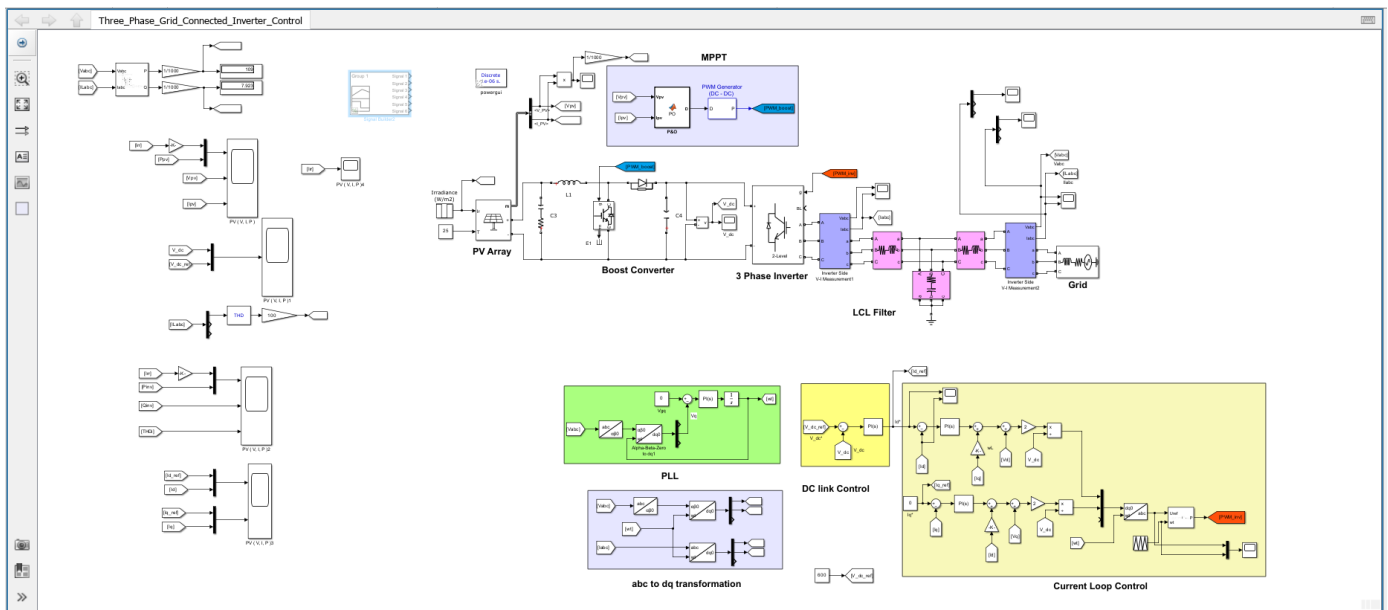


Figure 01: Complete Simulink model – Three-Phase Grid-Connected Inverter with MPPT

3. PERFORMANCE EVALUATION

PV Array Characteristics (I-V & P-V Curves)

The PV array used in the simulation is the REC Solar REC215PE (BLK) module configured as 11 series modules and 46 parallel strings. The I-V and P-V characteristics at 1000 W/m² for two operating temperatures (25°C and 45°C) are shown below.

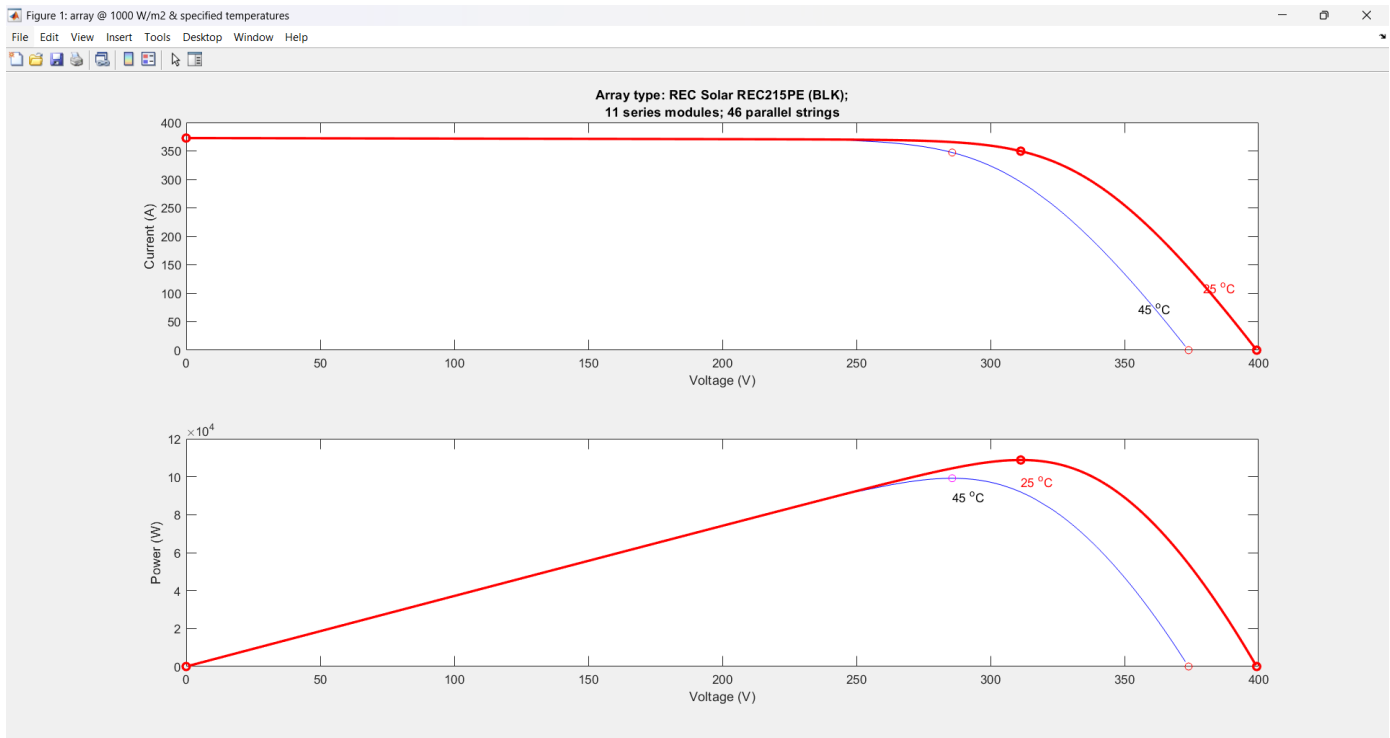


Figure 02: PV Array I-V and P-V characteristics at 1000 W/m² – 25°C and 45°C .MPP shifts with temperature.

- At 25°C, the array operates at a higher MPP voltage (~310 V) and peak power (~110 kW) compared to 45°C.
- Increasing temperature reduces open-circuit voltage (Voc) and shifts the MPP to a lower voltage, confirming the need for dynamic MPPT tracking.
- The P&O MPPT algorithm tracks these shifts continuously to extract maximum available power.

Irradiance Profile Applied

The irradiance profile applied during simulation consists of step changes to evaluate system response under dynamic conditions. The profile steps from 1000 W/m² down to 500 W/m² at $t = 0.2$ s, and recovers to 1000 W/m² at $t = 0.3$ s.



Figure 03: Applied irradiance profile

PV Array Output – Voltage, Current, and Power

The PV array output voltage (V_{PV}), current (I_{PV}), and power are shown in response to the irradiance step profile. All three quantities track the irradiance changes, with the MPPT algorithm maintaining operation near the maximum power point.

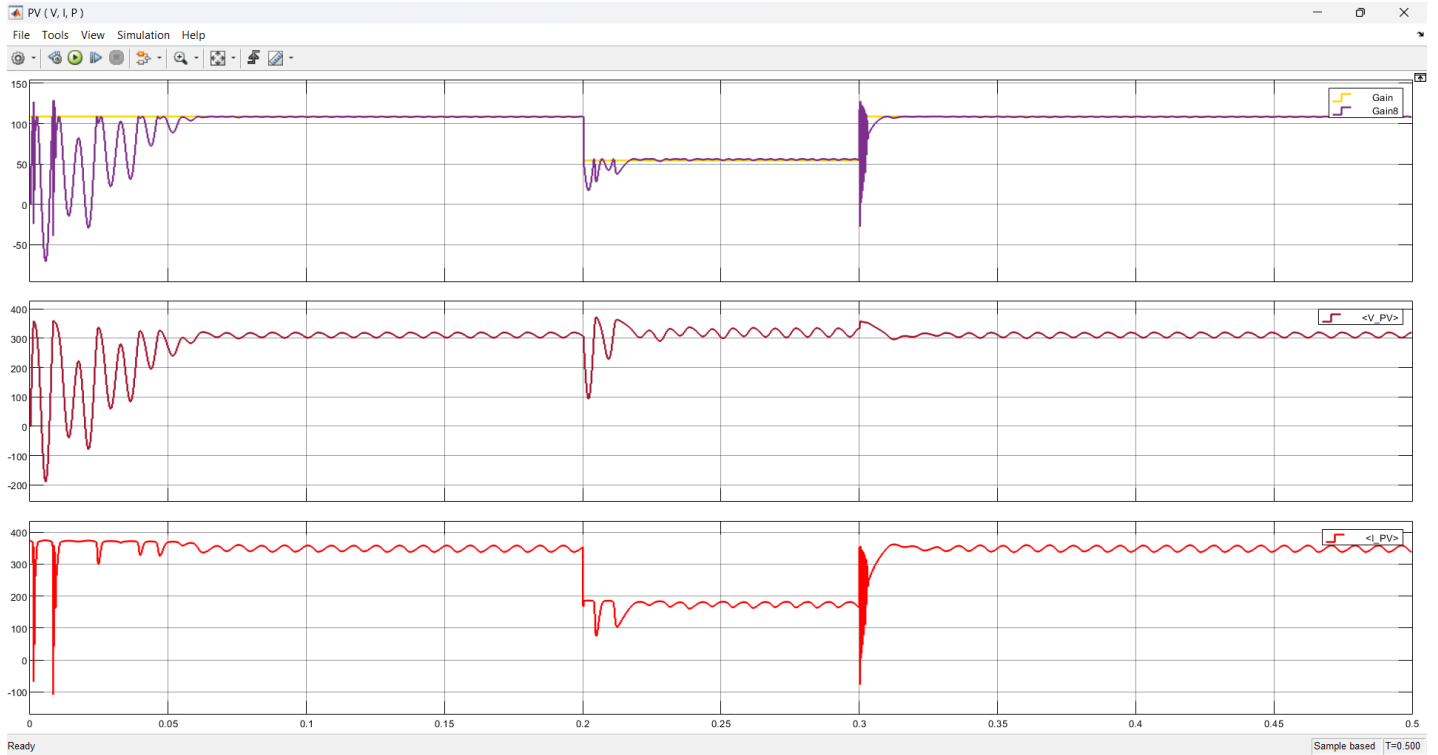


Figure 04: PV array output: Power, Voltage V_{PV} , and Current I_{PV} under the irradiance step profile

- During steady state at 1000 W/m^2 , power is maintained at approximately 105 kW with V_{PV} stabilised near 300 V and I_{PV} near 350 A.
- At the irradiance step-down ($t = 0.2 \text{ s}$), power drops to approximately 50 kW; the MPPT re-converges to the new MPP within approximately 50 ms.
- At the recovery step ($t = 0.3 \text{ s}$), the system rapidly returns to the original operating point, confirming fast dynamic response.

DC-Link Voltage Regulation

Two scope outputs show the DC-link voltage (V_{dc}) regulation performance. The outer PI voltage control loop maintains V_{dc} at the reference of 600 V under the irradiance step changes.

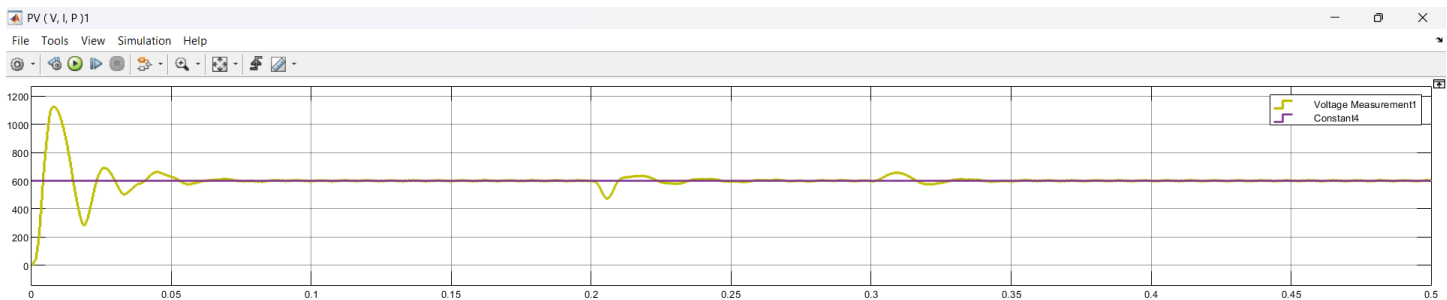


Figure 05: DC-link voltage tracks reference closely under irradiance steps

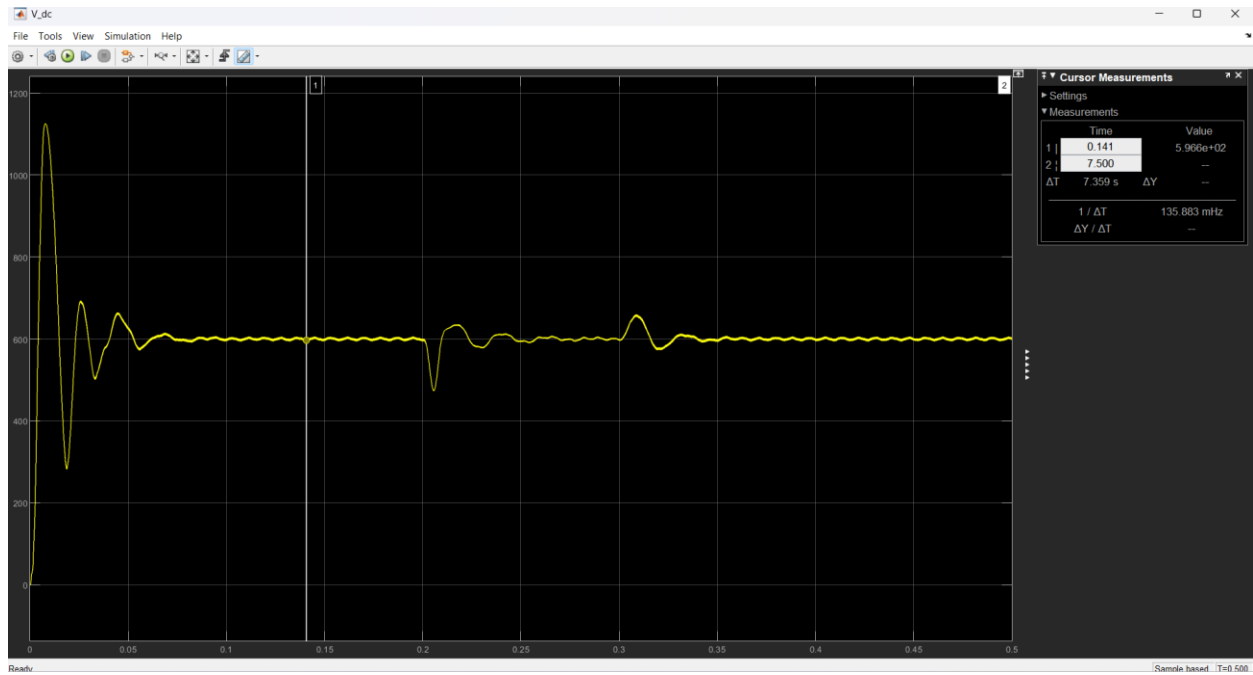


Figure 06: V_{dc} waveform

- V_{dc} settles to 600 V after an initial transient within approximately 50 ms from startup.
- At the irradiance step-down ($t = 0.2$ s), V_{dc} dips to approximately 480 V before recovering to 600 V within ~ 30 ms.
- At the irradiance recovery ($t = 0.3$ s), a brief overshoot to ~ 720 V is observed, settling back to 600 V within ~ 25 ms.
- The cursor measurement confirms steady-state $V_{dc} = 596.6$ V, representing a deviation of only 0.57% from the 600 V reference.

d-axis and q-axis Current Tracking (CLC)

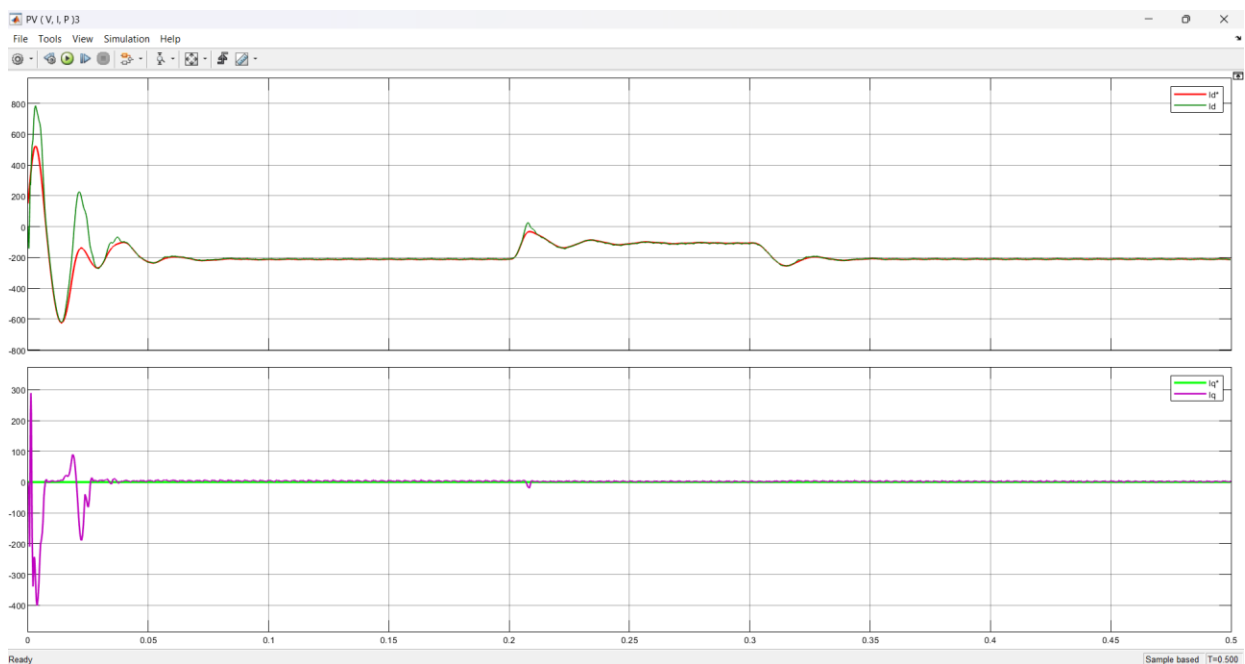


Figure 07: dq current tracking: I_d vs I_d^* and I_q vs I_q^* — CLC maintains tight reference tracking

The dq-frame current control performance is shown below. The d-axis current (I_d) tracks its reference (I_d^*) from the DC-link outer loop, while the q-axis current (I_q) is regulated to zero for unity power factor operation

- I_d tracks I_d^* closely in steady state; transient deviations during irradiance steps are quickly corrected by the PI regulators.
- I_q is regulated to zero ($I_q^* = 0$) confirming unity power factor operation — no reactive power injection into the grid.
- The CLC settling time after each irradiance step is within 20–30 ms, consistent with the inner loop bandwidth design.

PWM Modulation Signal

Scope 4 confirms the PWM modulation signal generated by the current controller output, which drives the SVPWM block for the three-phase inverter.

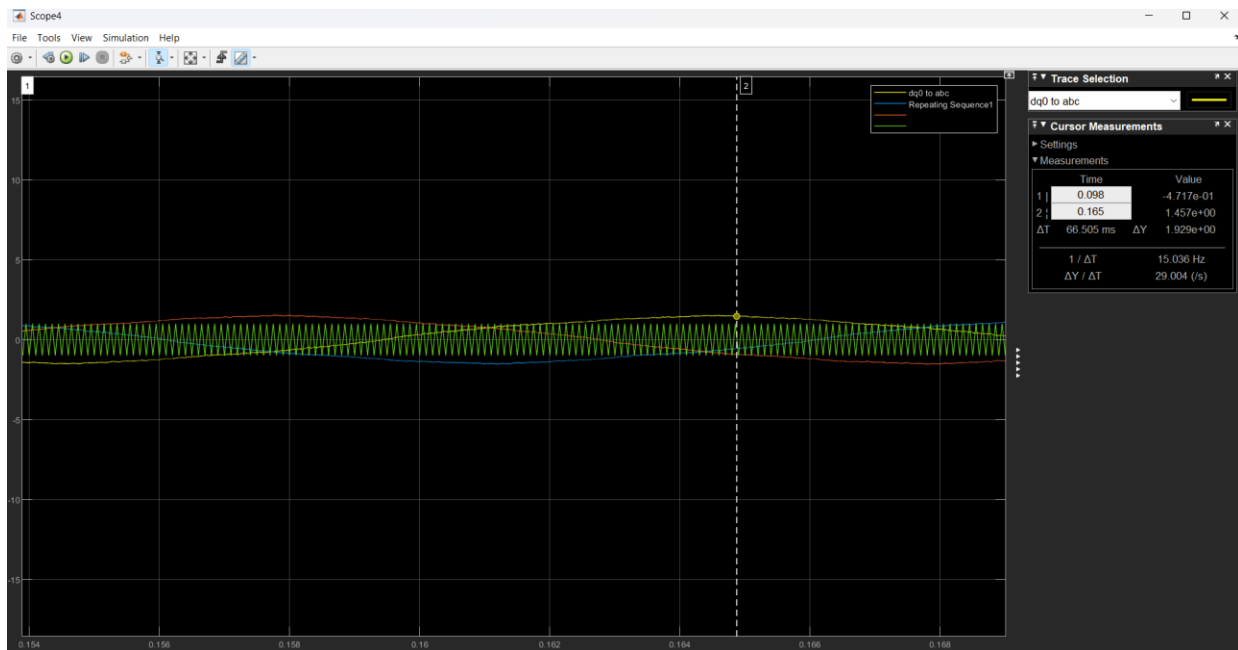


Figure 08: Scope 4: dq0-to-abc transformation output and PWM repeating sequence — modulation signal for SVPWM

- The modulation signal closely follows the repeating sequence reference, confirming correct dq0-to-abc back-transformation.
- The signal amplitude and frequency are consistent with 50 Hz grid synchronization provided by the PLL.

Inverter Output – Three-Phase Voltage and Current

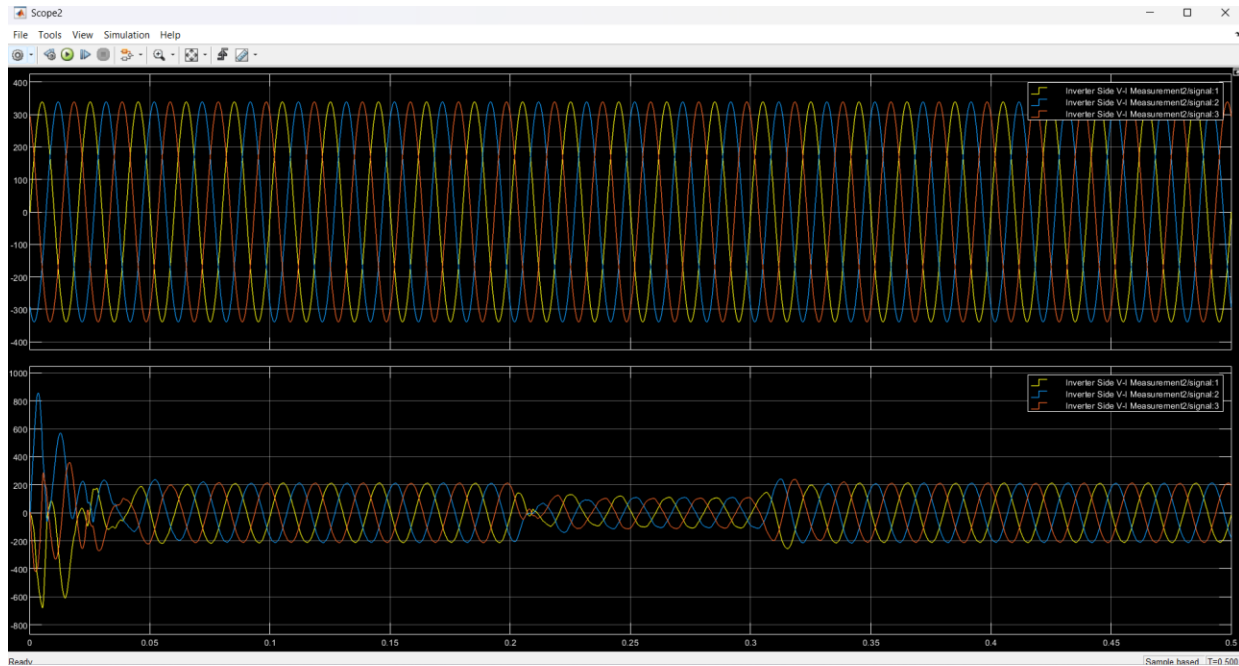


Figure 09: Three-phase voltage and current at grid side – clean sinusoidal waveforms after LCL filter

- After the LCL filter (Scope 2), the voltage waveform is clean and sinusoidal with amplitude approximately 311 V peak (220 V RMS), matching the grid voltage.
- Grid currents are balanced three-phase sinusoids, confirming correct phase synchronisation by the PLL.
- Current amplitude reduces at the irradiance step ($t = 0.2$ s) and recovers at $t = 0.3$ s, consistent with PV power changes.

4. Power Quality Analysis – THD Measurement

Perform a thorough Total Harmonic Distortion (THD) analysis on the grid current waveform to verify compliance with the harmonic distortion limit of $\text{THD} < 5\%$, as specified by IEEE 1547-2018 and IEC 61727 standards.

Methodology

THD analysis was performed using the MATLAB/Simulink Powergui FFT Analyzer tool on the steady-state grid current waveform. The analysis was configured with a start time of 0.28 s (after all transients have settled), a window of 1 cycle at 50 Hz, and the frequency axis displayed up to 1000 Hz (Nyquist frequency as max THD frequency).

FFT Analysis Result

The FFT Analyzer output from the simulation is shown below. The bar chart displays the harmonic spectrum (magnitude as % of fundamental) up to 1000 Hz, with the selected signal and FFT window shown in the upper plot.

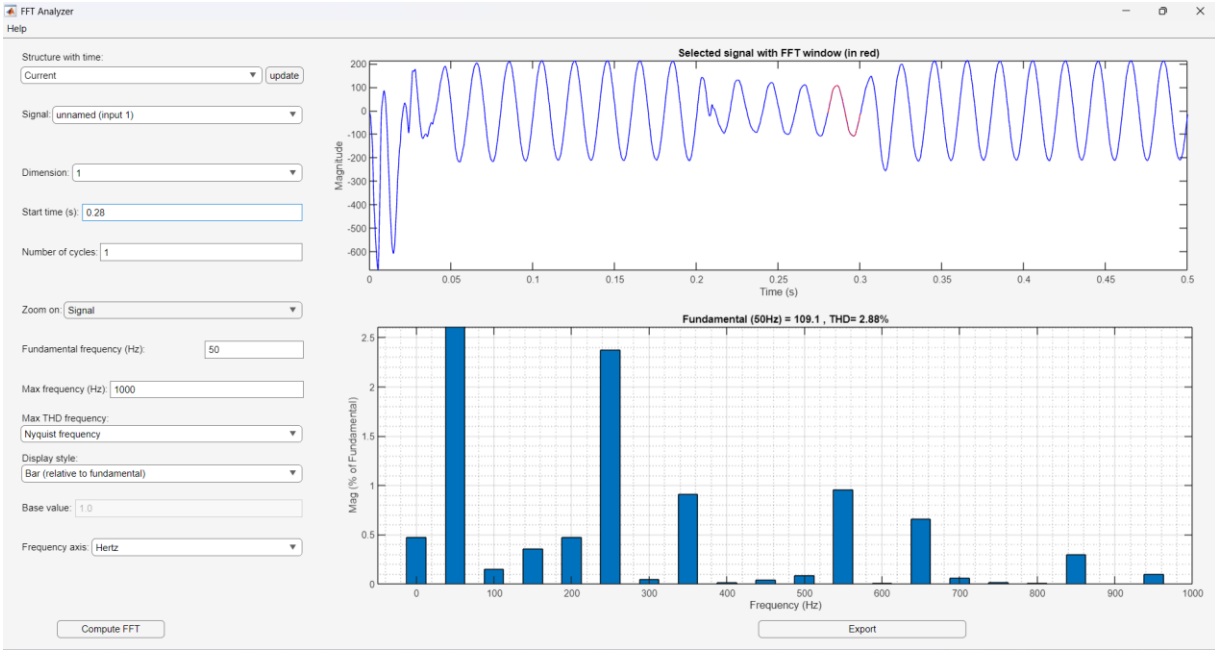


Figure 10: FFT Analyzer output

Measured Result: Fundamental (50 Hz) = 109.1 A | THD = 2.88%

- THD = 2.88% is well within the IEEE 1547-2018 limit of THD < 5%
- The dominant harmonics are concentrated in the low-frequency range (0–300 Hz)
- The 5th harmonic (250 Hz) shows the highest relative magnitude at approximately 2.4%
- Harmonics above 400 Hz are minimal, confirming effective LCL filter attenuation

Harmonic Spectrum Analysis

From the FFT bar chart, the following harmonic components are identified and compared against IEEE 1547 individual harmonic limits:

Table 01: Harmonic Spectrum Analysis and IEEE 1547 Compliance Results

Harmonic Order	Frequency (Hz)	Approx. Magnitude (% of fund.)	IEEE 1547 Limit	Result
3rd	150	~0.5%	< 4.0%	PASS
5th	250	~2.4%	< 4.0%	PASS
7th	350	~0.4%	< 4.0%	PASS
9th	450	~0.05%	< 4.0%	PASS
11th	550	~0.9%	< 2.0%	PASS
13th	600	~0.6%	< 2.0%	PASS
17th	700	~0.7%	< 1.5%	PASS
19th	800	~0.3%	< 1.5%	PASS
23rd	900	~0.1%	< 0.6%	PASS
25th	1000	~0.1%	< 0.6%	PASS

5. Future Plan

- ✓ Final Report Preparation – Document system architecture, MPPT algorithm implementation, PLL & CLC integration, THD measurement, efficiency comparisons, and stability assessment.
- ✓ Presentation Slides – Prepare slides summarizing project overview, proposed architecture, MPPT algorithm comparisons, simulation results, THD analysis, and conclusions.

6. Week 14 Summary

Table 02: Progress and Status

Topic	Description	Status
PLL Integration	PLL implemented for precise phase & frequency locking using dq-frame transformation.	Completed
CLC Implementation	dq-frame current loop control with PI regulators tested; inverter current tracking validated.	Completed
DC-Link Control	Outer voltage PI loop cascaded with CLC implemented; Vdc regulation at 600 V verified under varying irradiance.	Completed
MPPT Enhancement	Variable step P&O with global search phase implemented and tested under shading scenarios.	Completed
Efficiency Validation	Comprehensive efficiency testing across irradiance and temperature profiles planned for Week 14.	Completed
THD Analysis	FFT-based harmonic analysis targeting grid compliance (THD < 5%) planned for Week 14.	Completed
Documentation	Final report and presentation slides for Week 15 submission – drafting to begin.	Planned